Policy & Strategy Lab

Spine Case: Madhyapur

A stylised case used in every demonstration of the course. Madhyapur is not a real city. Its institutions follow the real structure of Indian urban governance, and the rules and schemes it refers to are real, but its figures, contracts and records are illustrative. They must never be cited as data about any real place.

Why a spine case

Every anchor tool in the Lab is demonstrated on the same case, so that participants see one complete project move from a vague concern to a defended recommendation. Framework-by-framework teaching makes it hard to see how the tools connect; a single running case shows it. The finished case also serves as the exemplar portfolio for the cohort.

The city

Madhyapur is a municipal corporation of about a million people, with residential wards of four broad types: planned colonies, older core wards, informal settlements and new peripheral wards. Household waste is collected door to door by a private contractor under a contract that pays by the tonne collected. Most collection vehicles have a single compartment. The city's landfill site is close to capacity. There is a composting unit working at capacity and no facility for dry waste. Informal waste pickers recover some recyclables from mixed waste and from the landfill.

The real framework around it

- The Solid Waste Management Rules, 2016 place a duty on waste generators, including households, to segregate waste at source into separate streams and hand it over to authorised collectors.
- Solid waste management appears in the Twelfth Schedule to the Constitution (item 6), added by the 74th Amendment, as a function states may devolve to municipalities. Public health and sanitation (State List, entry 6) and local government (State List, entry 5) are State List subjects.
- The Ministry of Housing and Urban Affairs runs Swachh Bharat Mission (Urban) 2.0, and Finance Commission grants to urban local bodies are partly tied to sanitation and waste management. Participants should check current guidelines before relying on any detail.
- Cities are ranked annually in the Swachh Survekshan survey.

The problem and the decision

Most Madhyapur households hand over mixed waste, contrary to the 2016 rules. The Municipal Commissioner must decide, before the next budget, which of three compliance strategies to fund in the coming financial year. The collection contract is due for renewal next year.

Where the case ends up

The analysis finds that separation fails mainly at collection: households stop separating when they see crews tip separated bags together, and crews mix waste because the contract pays for weight. Distributing bins alone barely changed behaviour, and fines without monitoring raised roadside dumping. The recommendation is a sequence: partner with a waste pickers' cooperative for separated dry-waste collection and ward sorting centres in ten pilot wards now, including informal settlements; then pay for separated collection when the main contract is renewed. The recommendation depends most on the cooperative's capacity to scale, which the pilot is designed to test.

How to use the case

- Demonstrate each day's anchor tool on Madhyapur before participants apply it to their own cases.
- Use the case to model analytical habits: the illustrative label on every figure, the Analyst's Ledger tags, the sealed Day 3 answer, and the honest verdicts on Day 14 (one hypothesis weakened, one rival supported).
- Point out where the case changes its mind. The sealed answer survives, but the diagnosis absorbs a rival explanation (vehicle compartments), and the recommendation becomes a sequence rather than a single choice.

The case, day by day

These are the worked examples used in each worksheet. Together they form the exemplar portfolio participants can consult: one complete project, from a vague concern to a defended recommendation.

Day 1: Problems, clients and decisions

Worksheet 1.1, Sorting your candidate cases. Anchor tool: Problem, symptom, cause and solution sort.

Madhyapur. Candidate: waste segregation. Statements: most households hand over mixed waste (problem); the landfill is close to full (symptom); crews tip separated bags into one compartment (candidate cause); give every household two bins (solution). Problem statement: most households in Madhyapur's residential wards do not hand over waste separated into the streams the 2016 rules require. Role: Municipal Commissioner. Decision: which compliance strategy to fund next financial year.

Day 2: Defining the problem

Worksheet 1.2, Problem Definition Canvas and Case Approval Form. Anchor tool: Problem Definition Canvas.

See the Madhyapur canvas on the Day 2 slides. Problem statement: most households in Madhyapur's residential wards hand over waste unseparated, contrary to the 2016 rules; the Municipal Commissioner must decide which compliance strategy to fund next financial year.

Day 3: Structuring with issue trees

Worksheet 2.1, Issue tree and sealed working answer. Anchor tool: Issue tree (MECE).

Madhyapur root question: why do most households hand over mixed waste? Level one, split by stage: households do not separate; collection mixes separated waste; separated waste has nowhere to go;

rules are not enforced or rewarded. Each branch ends in two or three answerable questions (see the Day 3 slides).

Day 4: Finding causes

Worksheet 2.2, Problem tree for your case. Anchor tool: Problem tree.

Madhyapur core problem: most households hand over mixed waste. Proximate cause: separating seems pointless, because households see crews tip separated bags together. Structural cause: the contract pays by the tonne, not by stream. Effects: a near-full landfill, lost recyclables, unsafe work.

Day 5: Systems and scope lock

Worksheet 2.3, Causal loop and the Gate 1 diagnostic. Anchor tool: Causal loop diagram.

Madhyapur loop: crews mix separated waste, households see the mixing, households stop separating, more mixed waste reaches crews, and so on (reinforcing). Obvious fix: fines. Unintended effects: roadside dumping; a burden on small homes.

Day 6: Jurisdiction, money and delivery

Worksheet 3.1, Jurisdiction, funding and delivery chain. Anchor tool: Delivery chain map.

Madhyapur: solid waste management is a Twelfth Schedule function; public health and sanitation and local government are State List subjects. Funding: the mission's central and state shares, Finance Commission grants to urban local bodies and the corporation's own revenue. Chain: Union ministry, state department, corporation, ward staff, contractor crews, households. Likely break: between ward staff and crews.

Day 7: Actors and political economy

Worksheet 3.2, Stakeholder and incentive matrix. Anchor tool: Power-interest map with absent voices.

Madhyapur: the Commissioner, councillors and the contractor are high power and high interest; waste pickers, sanitation workers and residents near the landfill are high interest with little voice. Rule on paper: households must separate. Rule in use: the truck takes everything.

Day 8: Prioritising issues

Worksheet 4.1, Pruning matrix. Anchor tool: Pruning matrix: size of effect against ability to influence.

Madhyapur: analyse first, collection mixing and processing capacity; keep, narrowed, household space and know-how; quick check, monitoring; prune to context, national recycling prices.

Day 9: Hypotheses

Worksheet 4.2, Hypothesis register. Anchor tool: Hypothesis register.

Madhyapur H1: payment by weight leads crews to mix separated waste. Weakened if crews mix just as much under per-household or per-trip contracts. Rival: vehicles lack compartments whatever the payment. Discriminating evidence: cities with compartment vehicles but weight-based contracts.

Day 10: The workplan

Worksheet 4.3, Workplan and evidence register. Anchor tool: Workplan matrix: hypothesis, question, analysis, source, deadline.

Madhyapur: H1 contract, compare cities with different payment terms using published contracts, audits and case studies, by Day 13; H3 processing, set capacity against waste generated using the city's waste plan, by Day 12.

Day 11: Finding and judging evidence

Worksheet 5.1, Building the evidence register. Anchor tool: Source reliability assessment.

Madhyapur register entries: the 2016 rules (law, high); the city waste plan (official plan, medium, states intentions); ward returns (administrative, low to medium, self-reported); a state CAG audit of waste management (audit, high for its period).

Day 12: What the numbers say

Worksheet 5.2, Data analysis. Anchor tool: Disaggregation and comparison in a spreadsheet.

Madhyapur (illustrative): separation by ward group, last year and this year. Finding: a sharp rise in planned colonies, little change elsewhere, incomplete data for informal settlements. Limitation: ward returns are self-reported.

Day 13: Experience and comparison

Worksheet 5.3, Comparative and qualitative evidence. Anchor tool: Comparative case matrix.

Madhyapur compared with Indore, Pune and Ambikapur: the principle of paying for separated collection and partnering with organised waste pickers may transfer; Indore's scale of spending and sustained leadership may not.

Day 14: Testing the hypotheses

Worksheet 6.1, Hypothesis testing matrix. Anchor tool: Hypothesis testing matrix.

Madhyapur (illustrative): H1, the contract, supported with the vehicle rival partly true; H2, containers, weakened; H3, processing, partly supported; H4, enforcement, the rival supported: fines alone backfire.

Day 15: Synthesis and the diagnostic memo

Worksheet 6.2, Diagnostic memo. Anchor tool: Pyramid Principle with SCQA.

Madhyapur governing message: separation fails mainly at collection, because the contract pays for weight; campaigns alone will not fix it. Key lines: households respond to what crews do; crews respond to what they are paid for; downstream capacity decides whether separation is worthwhile.

Day 16: Generating options

Worksheet 7.1, Options canvas. Anchor tool: How Might We questions with a journey map.

Madhyapur: baseline (current contract and campaigns); A, pay for separated collection; B, partner with a waste pickers' cooperative for dry waste; C, a ward enforcement drive. A and B change instruments; C adjusts settings.

Day 17: Choosing between options

Worksheet 7.2, Options appraisal. Anchor tool: Options appraisal matrix with a distributional row.

Madhyapur (illustrative): B scores 3.35, the baseline 3.15, A 3.00 and C 2.30. B is robust to the weights but sensitive to the cooperative's reach. Recommendation: B in ten pilot wards now, moving to A at contract renewal. Weakest leg: operational capacity.

Day 18: Designing for delivery

Worksheet 8.1, Implementation plan. Anchor tool: Theory of change.

Madhyapur option B: inputs (funding, cooperative contract, sorting centres) to activities (pickers collect dry waste) to outputs (separate dry waste collection in pilot wards) to outcomes (households separate more; less to landfill). Key assumption: households respond to seeing separation respected. Approach: adaptive, ten-ward pilot with quarterly review.

Day 19: Measurement and the pitch

Worksheet 9.1, Indicator framework and draft deck. Anchor tool: Indicator framework (indicator tree).

Madhyapur option B: outcome indicators (households separating by ward type; dry waste reaching landfill), output indicators (dry waste collected by the cooperative; pickers with contracts), and an early warning on the key assumption (separation in pilot wards after three months).

Day 20: The defence

Worksheet 10.1, Capstone submission. Anchor tool: Panel presentation and defence.

Madhyapur: the panel's finance officer asks what happens if the cooperative reaches only half the wards. The report adds the sensitivity result (cost per household roughly doubles) and a decision rule at the first quarterly review.